

SPC project in R: readme

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I made this project primarily to further my R knowledge as I know I will get to use it again in the future (hopefully within the business analytics master). It also helped me further understand SPC and the statistics behind it. I will thereafter try to briefly present how to use the tools of this project.

While there are many functions that can be used on their own, they are bundled and integrated within 2 functions to facilitate usage, one handles phase 1 and the other phase 2:

- **SPC_Phase1**(data, lsl = NULL, usl = NULL, type = "auto", target = 1.33):
 - data:
For process characteristics: a matrix with the rows being the number of subgroups and the column being the size of subgroups.
For process attributes: a matrix with the first column being the number of defects or defectives and the second column being the size of area of opportunities or sample sizes (they don't need to be constant).
 - lsl: lower specification limit
 - usl: upper specification limits, if process limits aren't set the function will just assess stability
 - type: "I-MR", "Xbar-R", "Xbar-S", "c", "u", "p", "np"
"auto" determines relevant type among "I-MR", "Xbar-R" and "Xbar-S"
"defects" determines relevant type among "c" and "u"
"defectives" determines relevant type among "p" and "np"
 - target: threshold Cpk for the process to be considered capable

This functions compute and returns the following values and informations in a list (list of 2 lists if it is a dual chart): Chart type, Lower control limit, Center line, Upper control limit, Individual standard deviation, Number of observations, Size of subgroups and values plotted. It also displays the control chart(s) with a sentence at the bottom stating stability and capability if specifications were provided.

- **SPC_Phase2**(CL_output,data,type="auto"):
 - CL_output: outputted list by the SPC_Phase1 function (it is named this way because the subfunctions for building that list is CL, for control limits)
 - data: same as **SPC_Phase1**
 - type: same as **SPC_Phase1**

This function computes the same list as in **SPC_Phase1** but it uses the control limits from the provided list. It also plots the control chart(s), stating stability at the bottom, it does not compute capability.

A process is considered stable here when all values are within control limits and if there are no cases of oscillation or trends.

Process capability computation is based on Cpk, which assumes normality, as such capability is never checked for c,u,p and np chart types.

Within the testing environment there are functions to create matrices of observations: **gen_variable** for normally distributed observations, **gen_defects** for Poisson distributed defects observations and **gen_defectives** for binomially distributed defectives observations. The last 2 accept a vector of n for variable sample size.

AI usage statement: I used Gemini as a helper throughout the creation process, I however refrained from simply copying its output, I instead prompted it in such a way that it could give me “hints” on how to attain my objectives. This really helped me to learn R and to strengthen my problem-solving skills while avoiding overreliance on AI to code.